

Toward Sustainable O-RAN Deployment: An In-Depth Analysis of Power Consumption

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Abstract—5G delivers limitless mobile broadband capabilities, enabling extensive connectivity for various devices, including smartphones, sensors, and machines. Its most notable feature lies in its ability to facilitate instantaneous and highly reliable machine communications. Because of the rapid evolution of communication networks, innovative architectures such as Open RAN (O-RAN) have emerged, promising increased flexibility, efficiency, and sustainability. To realize the full potential of O-RAN networks, however, it is critical to understand their power consumption implications. In this paper, in order to understand the way O-RAN consumes energy, a comprehensive power consumption model is used. The study not only provides theoretical analysis but also incorporates a practical implementation of the architecture. For this, an algorithm is employed to determine a base monolithic architecture of O-RAN by the placement of Distributed Unit (DU) and Centralized Unit (CU) using available data of Radio Unit (RU) in France and Ireland. The research further explores the impact of different parameter variations in various Evaluation Scenarios (ESs) on the architecture. By conducting this thorough evaluation, this research provides valuable insights for researchers and industry practitioners to gain a deeper understanding of energy consumption patterns and derive ideas on how to effectively reduce power consumption in these networks.

Index Terms—Green networking, cloud, open-ran, 3-layer architecture, mobile, algorithm, placement.

I. INTRODUCTION

OVER the last two decades, cellular networks have evolved to accommodate technological innovations from Second-Generation (2G) to Fourth-Generation (4G). They are now gradually shifting to Fifth-Generation (5G), intending to meet the diverse needs of various industries. The 5G

Core (5GC) supports both core and access networks through programmability, network slicing, and a service-based architecture (SBA) [1]. 5G provides unlimited mobile broadband, massive device and machine connectivity, and dependable machine communications [2]. Developers are making efforts to develop network capabilities as 5G becomes a reality. Telecom operators and vendors are working together to develop independent and interoperable components, with a focus on software-driven, intelligent, flexible, and energy-efficient solutions [3], [4].

The Radio Access Network (RAN) stands as a promising domain, offering significant advantages through the integration of hardware and software components independently. This fosters increased competition within the ecosystem and empowers network operators to select the most suitable solutions for their specific networks. As a result, network architectures are shifting away from conventional closed structures towards open radio networks, often referred to as Open RAN, O-RAN [5].

O-RAN, an industry-wide standard for RAN interfaces, enables equipment from different vendors to work together, increasing network flexibility at a lower cost. O-RAN's main goal is to achieve interoperability among RAN elements through standardized communication interfaces, regardless of the hardware and software components used by different vendors. This allows for more openness and innovation in the RAN market, as well as lower entry barriers for new players.

In parallel, a key element that enables the development of open and flexible radio access networks is the adoption of softwarization principles in O-RAN architectures [6]. Softwarization principles refer to the use of software-based solutions to implement network functions and services, instead of relying on dedicated hardware devices. By applying softwarization principles, O-RAN architectures can achieve higher scalability, flexibility, and programmability, as well as lower costs and faster innovation. Although softwarization has advantages like increased flexibility and efficiency, it also poses sustainability challenges, particularly in terms of increased energy consumption due to increased computational needs and data processing in software-based networks [7].

In O-RAN, the substantial energy consumption linked to the access component represents another issue with sustainability. Base stations and related infrastructure that are part of access networks can use up to 50% of the total energy consumed by the network segment [8]. The high energy consumption of O-RAN networks in the access network segment raises concerns

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over their effect on the environment and carbon footprint. It emphasizes the need for energy-efficient designs, technologies, and operational procedures in order to reduce energy consumption and promote sustainability in these networks.

In this paper, we look into O-RAN's power consumption with a focus on energy efficiency in order to assess its impact on the sustainability of today's and upcoming networks. To accomplish this, we propose a thorough model to evaluate the energy consumption of O-RAN under various circumstances. The model allows for evaluating the architecture's energy efficiency and takes into account real deployment scenarios. Results from the model's application to a real deployment show the energy consumption trends and offer important information for improving power efficiency in O-RAN networks.

The evaluation provided in this study is based on the infrastructure that is currently available, and while it offers insightful information in the context of the existing conditions, it also acts as a worst-case scenario. This worst-case scenario offers a significant benefit by highlighting the elements that need to be carefully taken into account in order to achieve a truly sustainable deployment in the near future. We can proactively address potential obstacles and clear the way for a more sustainable O-RAN deployment in the future by carefully examining the opportunities and challenges mentioned in this evaluation. Although our assessment is based on the current state of technology, the same principles and approaches can be used to foresee potential advancements and challenges in the O-RAN ecosystem in the future.

The remaining sections of the paper are organized as follows: Section II provides a comprehensive overview of the state of the art in O-RAN networks. Section III presents a parametric model of power consumption, which encompasses the evaluation of energy usage in the O-RAN framework. Additionally, this section discusses the process of determining a practical placement of edge facilities within the network architecture. Section IV focuses on the presentation and analysis of numerical results obtained from applying the proposed power consumption model to real-world scenarios. Lastly, Section V provides the conclusions of this paper.

II. STATE OF THE ART

In this section, we give a thorough overview of the state-of-the-art O-RAN architectures, O-RAN network sustainability issues, and the current status of O-RAN deployments. We go into each of these factors to establish the groundwork for understanding the advancements, challenges, and potential opportunities in the quickly expanding O-RAN technological landscape.

A. Architecture

A novel design for O-RAN has recently been proposed by the 3rd Generation Partnership Project (3GPP) and is based on the division of baseband functions into three entities or “*splits*” [9]: Centralized Unit (CU), Distributed Unit (DU), and Radio Unit (RU). Higher-layer processing is often handled by CUs, which are typically located at higher-hierarchy metro

nodes, such as a sizable Central Office (CO), whereas lower-layer processing is handled by DUs, which are distributed closer to the antennas (RUs) [10].

The 3-layer architecture enables a more flexible deployment of baseband operations in comparison to the conventional 2-layer architecture, where processing functions are only spread between two elements: Remote Radio Head (RRH) and Baseband Unit (BBU) [11]. Additionally, it is anticipated that the positioning of CUs and DUs strategically will result in reduced additional costs, power consumption and service blocking [12]. It is challenging to accomplish these goals while yet adhering to service latency standards.

Centralising baseband operations decreases energy consumption but significantly raises the demand for transport capacity. On the other hand, a more dispersed placement of CUs and DUs results in excessive energy usage, while allowing for a reduction in the amount of network capacity needed. In order to jointly meet capacity, latency, and energy needs, it is crucial to take advantage of the flexibility provided by the 3-layer design.

Recent research has already looked into the DU/CU placement dilemma; however, the majority of contributions focused on a static implementation. With the aim of lowering network power consumption, [13] modeled the CU placement problem using an Integer Linear Program (ILP) and took into account several split alternatives. Similar to this, [14] proposed an ILP to reduce the amount of processing power needed to support 5G network services, while still meeting the latency requirements of selected splits.

B. Sustainability Issues

When we look at the challenges, one of the most critical concerns is the sustainability aspect, particularly regarding power consumption. Several telecommunications corporations have established carbon neutrality targets for the next 10 to 15 years. The trials and testing that employ Commercial Off-the-Shelf (COTS) servers consume at least three to four times (if not more) the power of a standard RAN system [15]. A rollout with O-RAN, especially now, would make it much harder to meet the carbon neutrality goals.

The good news is that chip makers are inventing and designing chipsets that are specifically intended for 5G physical layer performance while also keeping the programmability required by O-RAN. Such developments will help to optimize the power consumption of O-RAN equipment like DUs and CUs. Another option for power savings exists in the O-RAN centralized method. When we use lower layer splits in O-RAN, most of the processing is performed in a centralized location, where processing can be scaled up or down depending on the traffic requirements, benefitting from cloudification. This would save energy costs as compared to having distributed processing capabilities at each base station.

In the past decade, to address such a sustainability goal, several projects focused on Energy Efficiency (EE) in mobile networks; among others, we can mention: EARTH [16], GreenTouch [17], OPERA-Net 2 [18], 5GrEEn [19], and ECONET [20].

The overall target of the EARTH project was to reduce the power consumption of mobile networks with preserved Quality of Service (QoS) [16]. It contributed to the understanding of the energy consumption breakdown in wireless access networks by answering the key question: “*how much energy is needed to run a wireless network?*” [21]. It was one of the first collaborative efforts by large telecommunications vendors and operators to reach an agreement on possible strategies to improve existing networks’ efficiency in terms of energy consumption.

GreenTouch followed EARTH as a very large network of industrial and academic forces and was dedicated to reducing the carbon footprint of Information and Communication Technologies (ICT) devices, platforms, and networks. Their mission was to deliver the architecture, specifications, and roadmap, and had the ambition to increase the network EE by a factor of 1000 compared to 2010 levels [17].

Celtic-Plus OPERA-Net 2 (2011-2015) investigated energy-efficient wireless networks using a comprehensive strategy that included radio resource management algorithms, components, life cycle assessment, and innovative cooling technologies [18].

5GrEEn targeted the EE aspects of 5G mobile networks, and hence contributed to the important design target of low energy consumption in mobile networks [19]. The project presented load-adaptive massive Multiple-Input and Multiple-Output (MIMO) solutions [22], addressed the importance of backhaul on the mobile network energy consumption and proposed green mobile backhauling solutions [23]. 5GrEEn also focused on the decoupling of the data plane from the control plane [24] together with GreenTouch.

The ECONET project addressed specifically the green management and control of the fixed network segment. It targeted dynamic adaptation (e.g., adaptive power and low-power-idle) and smart sleeping techniques tailored for “traditional” networking devices (Ethernet interfaces, switches, routers,...) with no virtualization. A recent attempt to overcome some difficulties in power adaptation arising in virtualized environments and to extend the techniques introduced in ECONET to softwarized networks is being conducted in the framework of the Horizon Europe Program by the recently launched 6Green project.¹

Network infrastructure scalability is amongst the main challenges facing 5G deployments. For instance, to scale up network capacity per unit area as per the requested demands, the underlying radio access network needs to be massively densified compared to the current deployments [25]. From a number of angles, including but not limited to energy consumption, cost, operation and maintenance, delay, handovers, and others, such densification is challenging [26]. While such densification is inevitable, centralization and virtualization of resources can still offer adequate solutions, keeping all to-be-distributed nodes as simple, cost-effective, and energy-efficient as possible, especially if further equipped with smart energy-saving/sleeping features. Specifically, the rapid advancements and availability of energy-efficient and powerful centralized

computation offered by datacenter technology can help if computation clusters are formed on the access side.

For the goal of pooling a great number of cells, Virtual RAN (v-RAN) uses general-purpose hardware and the eNBs protocol stack in a virtualized environment [27]. Therefore, baseband resources can be centrally managed in a Centralized RAN (C-RAN) configuration. This is exactly the reason that significant research efforts have been recently focused on C-RAN technology. The main goal of C-RAN is to centralize all functions related to digital baseband computation and cooling at specific cluster centers in order to reduce costs and energy usage, leaving only Radio Frequency (RF) and analog processing at the cell sites. On top of the benefits, this centralization can also be an enabler for Coordinated MultiPoint (CoMP) technology, enabling millisecond-scale inter-cell coordination that is more rapid, leading to better resource utilization and EE, particularly for cell-edge users.

Although the C-RAN idea remains promising, several challenges remain to be tackled to practically apply it side-by-side with the arising technologies. Congestion of fronthaul links, connecting centralized nodes to distributed ones, is one consequence of the functional splitting suggested in C-RAN. Compared to its preceding Distributed RAN (D-RAN) architecture, full centralization of physical-layer processing, including channel coding, modulation, and Fast Fourier Transform (FFT), can significantly upscale the fronthaul signal bandwidth in several orders of magnitude, especially as the quantity of spatial streams and serving antennas both increase. That is, such an effect can become much more limiting when technologies like massive MIMO are employed. This problem can be found in both uplink and downlink directions. For instance, the fronthaul bandwidth also increases when uplink multi-user detection scenarios, such as CoMP joint reception, call for higher signal resolution (finer quantization). Therefore, more convenient, and potentially flexible, intermediate functional splits need to be further studied to both keep the scalability benefits of C-RAN, while resolving the bandwidth congestion and allowing for effective CoMP coordination for cell-edge users. We call this flexible semi-centralized architecture herein Hybrid C-RAN (H-CRAN), as proposed by SooGREEN in [28].

In the context of realizing a practical C-RAN, several research efforts have been exerted, which can be classified as follows: i) Energy and cost studies of fronthaul, ii) Processing allocation, and iii) Processing split [29]; Others have looked into the creation of a fronthaul protocol to transfer synchronized control and user data information from a central site housing the BBU pool to RUs (enabling processing at two sites) [30]; Analytical studies, including market sharing based on price quotations, on techniques to enhance overall C-RAN system’s capacity (radio and processing) can be found in [31].

C. Deployment

To achieve an environmentally-friendly network, network designers must take into account sustainability across a wide spectrum of factors. One aspect is to investigate whether the underlying network architecture and the deployed equipment

¹<https://www.6green.eu/>

enable efficient power saving based on the varying traffic load. Recent network deployments constitute heterogeneous cell coverages (*macro*, *micro*, *pico*, *femto*) that overlap on one another to support high traffic loads per unit area. From an EE point of view, it needs hence to be ensured that at any given time such a capacity enhancement is autonomously enabled only when needed. Also, optimizing the placement of computational and communication resources over the underlying infrastructure can be a major challenge in the overall network energy consumption, that will eventually affect the placement of powering and cooling facilities, and the transport link capacities between the different serving nodes, among others.

Nowadays, RANs are facing a rapid increase in traffic demand, due to the increasing number of devices connected to the network, along with the higher quality of service requested by the users. Network operators must be able to provide the required capacity to fulfill such demand. C-RAN has recently been presented as a promising design to address this problem by expanding network capacity, enhancing EE, and supplying scalability and flexibility. In a traditional D-RAN, the components constituting the Base Station (BS) are located at the cell site and are divided into a RU for the transmission and reception of radio signals, and a BBU for digital baseband signal processing.

The main idea of C-RAN is to centralize all BBUs (i.e., DUs), in a shared location responsible for coordinated signal processing and management, which is called BBU hotel, while connecting to the RUs at the cell sites, now also called RRHs, using optical fiber links. Such a centralization allows to share maintenance costs and energy consumption among several BBUs. C-RAN represents one further development to RAN where the processing load of several RRHs can be multiplexed into a smaller number of BBUs via pooling and virtualization, now with the BBU hotel turned into a BBU pool. Each RRH's digital baseband processing is specifically carried out by a Virtual Machine (VM), and multiple VMs can coexist on a single General-Purpose Processor (GPP) at the BBU pool. This idea can offer further energy saving and better network scalability. Despite its appealing EE and cost reduction features, additional network challenges are introduced by C-RAN, though. In particular, the transmission bottleneck arises due to the transmission delay and fronthaul capacity constraints between the Centralized Cloud (CC), where the BBU pool is hosted, and the intended RUs. This becomes more pronounced as the number of users and data rates served by the corresponding RU increase, mainly because fully processed high-bandwidth Radio-Over-Fiber (RoF) signals are now being transmitted from the BBU pool to the RUs.

However, C-RAN also has some limitations and drawbacks, such as the high cost and complexity of the fronthaul links, the lack of flexibility and interoperability among different vendors, and the security and privacy risks of centralizing the data processing. To address these challenges, O-RAN has surfaced as a novel approach seeking to unveil the RAN architecture and interfaces. Its objective is to foster greater intelligence and innovation within the network. O-RAN introduces the idea of three functional splits; RU, DU, and CU,

providing flexibility for varying degrees of centralization or decentralization based on network requirements and conditions [32]. O-RAN utilizes Artificial Intelligence (AI) and Machine Learning (ML) methods to enhance network performance and efficiency. It is designed to accommodate multi-vendor and multi-technology scenarios. O-RAN is anticipated to offer advantages surpassing those of C-RAN, including reduced costs, increased scalability, enhanced quality of service, and improved energy efficiency [33].

III. PROPOSED MODEL

This section is divided into two subsections, each focusing on distinct aspects of the analysis. The first subsection delves into the development of a comprehensive power consumption model. This model serves as a key component for evaluating the energy consumption of the O-RAN implementation. The second subsection centers around determining an optimal placement of edge facilities within the network architecture, by exploiting the metrics defined in the model. This step employs an efficient placement algorithm to achieve an optimal and minimal placement solution for DUs and CUs.

A. Parametric Model of Power Consumption

In this section, we describe our parametric model to represent the power consumption considering the O-RAN specification architecture with the goal to estimate the power difference with the case of a simple RAN architecture. In Table I, a detailed description of our notation appears.

The different physical elements which are part of the O-RAN and RAN architectures can be divided into three main components:

- Processing Unit (PU);
- interface; and
- Common Site Infrastructure (CSI).

The main part of power consumption in each node in these architectures is in the Processing Unit (PU). To calculate this power, we need to know about different processes in these nodes and how much they are dependent on the amount of processed traffic. These complexity values determine how many billion complex arithmetic operations are carried out each second and are expressed in Giga Operations Per Seconds (GOPS). Based on a model proposed in [34], the power consumption of a PU can be calculated as:

$$\tilde{\Phi} = \sum_{i \in I} f_i \left(\frac{C_i}{T_i} \right) \quad (1)$$

where $I = \{1, \dots, n\}$ is a set of different $n \in \mathbb{N}$ sub-components that constitute the PU, while $f_i(\cdot)$ is a function representative of the contribution of the sub-component $i \in I$ to the total power consumption. The function $f_i(\cdot)$ is dependent on the ratio between the reference complexity C_i of the i_{th} function (expressed in GOPS), and the technology-dependent parameter T_i that indicates the hardware complexity based on system configuration and on the date of system deployment

TABLE I
NOTATION USED IN THE PARAMETRIC POWER MODEL

Symbol	Description
$\bar{\Phi}$	PU power consumption.
$I = \{1, \dots, n\}$	Set of different $n \in \mathbb{N}$ sub-components.
$f_i(\cdot)$	Function representative of the contribution of the sub-component $i \in I$.
C_i	Reference complexity of the sub-component $i \in I$.
T_i	Technology-dependent parameter of the sub-component $i \in I$.
$\alpha_i > 0, \beta_i > 0, \gamma_i > 0$	Proportional, additive and scaling parameters of the impact of the sub-component $i \in I$ to the power consumption of the PU.
$\delta_{DU} \in [0, 1]$	Fraction of the sub-component mapping shared by DU and CU.
$\bar{\Phi}_{RRH}, \bar{\Phi}_{BBU}$	PU power consumption of RRH and BBU elements of the RAN architecture.
$\lambda \in [0, 1]$	Normalized incoming workload.
$\bar{\Phi}_{RU}, \bar{\Phi}_{DU}(d), \bar{\Phi}_{CU}(c, \lambda)$	PU power consumption of RU, $d \in \{1, \dots, n_{DU}\}$ DU, and $c \in \{1, \dots, n_{CU}\}$ CU elements of the O-RAN architecture.
$\psi_d^{(RU)}$	Number of RUs that are connected to the DU $d \in \{1, \dots, n_{DU}\}$.
$\psi_c^{(DU)}$	Number of DUs that are connected to the CU $c \in \{1, \dots, n_{CU}\}$.
$\omega_{CU}(c)$	Number of physical instances needed to satisfy the processing capacity of the CU $c \in \{1, \dots, n_{CU}\}$.
$\mu \in [0, 1]$	Maximum supported normalised capacity of the physical instance used to perform the operation of a logical entity.
$\bar{\Phi}_{CU}^{(act)}(c), \bar{\Phi}_{CU}^{(idle)}(c)$	PU power consumption of the CU $c \in \{1, \dots, n_{CU}\}$ during the active and idle periods.
Φ_{phy}	Power consumption of the chosen physical instances to implement the logical entities.
T_{phy}	Technology-dependent parameter of the chosen physical instances to implement the logical entities.
$s_{act} > 0$	Scaling parameter of $\bar{\Phi}_{CU}^{(act)}(c)$ with $c \in \{1, \dots, n_{CU}\}$.
$\theta_{idle} \in [0, 1]$	Ratio of the active power consumption in the idle periods to the total PU power consumption of the CU.
$\bar{\Phi}_{CU}(c)$	Expectation of $\bar{\Phi}_{CU}(c, \lambda)$ with $c \in \{1, \dots, n_{CU}\}$ with respect to the incoming workload λ .
$\rho_{\lambda}(\lambda)$	Probability density of the incoming workload.
$\Phi_{FH}, \Phi_{MH}, \Phi_{BH}$	Power consumption of the fronthaul, midhaul, and backhaul interfaces.
Φ_{CSI}	Power consumed for cooling, monitoring, and lights.
Φ_{RRH}, Φ_{BBU}	Power consumption of RRH, and BBU RAN elements.
$\Phi_{RU}, \Phi_{DU}, \Phi_{CU}$	Power consumption of RU, DU, and CU O-RAN elements.
Φ_{RAN}, Φ_{O-RAN}	Power consumption of the RAN and O-RAN architectures.
n_{RRH}, n_{BBU}	Number of RRH and BBU elements of RAN architecture considering a specific scenario.
n_{RU}, n_{DU}, n_{CU}	Number of RU, DU, and CU elements of O-RAN architecture considering a specific scenario.
$\Delta\Phi\%$	Power percentage difference of the O-RAN architecture compared to the RAN one.

(expressed in GOPS/W). We can express the function $f_i(\cdot)$ in the following way:

$$f_i\left(\frac{C_i}{T_i}\right) = \alpha_i \cdot \left(\frac{C_i}{T_i}\right)^{\gamma_i} + \beta_i \quad (2)$$

where the parameters $\alpha_i > 0$, $\beta_i > 0$, and $\gamma_i > 0$ represent the proportional, additive, and scaling parameters of the sub-component $i \in I$.

Table II shows the mapping of the different sub-components of the set I to the elements of the RAN and O-RAN architectures. For the RAN case, the sub-components with $i = \{1, \dots, 13\}$ are part of the RRH, while the last two represent the BBU. The sub-component with $i = 14$ represents the digital functions that perform platform control, such as activating different parts of the system and managing data flow, while the sub-component with $i = 15$ represents the digital functions that perform network processing, such as higher-layer protocols and backhauling to the core network. Instead, for the O-RAN specification, the RU is composed of the sub-components with $i = \{1, \dots, 12\}$ and a fraction of

TABLE II
SUB-COMPONENT MAPPING TO RAN AND O-RAN ARCHITECTURAL ELEMENTS AS DEFINED IN [34]

	Sub-component	RAN	O-RAN
1	Predistortion		
2	Filtering		RU
3	Up/Down-sampling		
4	TD estimation/compensation of non-idealities		
5	FFT/IFFT, FD non-idealities.		
6	MIMO precoding		
7	Synchronization		
8	Channel estimation & interpolation		
9	Equalizer computation		
10	Equalization		
11	OFDM Modulation/Demodulation		
12	Mapping/Demapping		
13	Channel Coding		
14	Platform Control		BBU
15	Network Processing		CU

the sub-component with $i = 13$, whereas the DU is composed of the fractions of sub-components with $i = \{13, 14\}$; finally, the CU is composed of the fraction of the sub-component with $i = 14$ and the one with $i = 15$.

Summarizing, the power consumption of the RAN elements can be expressed as follows:

$$\bar{\Phi}_{RRH} = \sum_{i=1}^{13} f_i\left(\frac{C_i}{T_i}\right) \quad (3)$$

$$\bar{\Phi}_{BBU} = \sum_{i=14}^{15} f_i\left(\frac{C_i}{T_i}\right) \quad (4)$$

Instead, the power consumption of the generic RU and of the DU elements $d \in \{1, \dots, n_{DU}\}$ can be expressed as:

$$\bar{\Phi}_{RU} = \sum_{i=1}^3 f_i\left(\frac{C_i}{T_i}\right) \quad (5)$$

$$\bar{\Phi}_{DU}(d) = \sum_{i=4}^{13} f_i\left(\frac{C_i \cdot \psi_d^{(RU)}}{T_i}\right) + \delta_{DU} f_{14}\left(\frac{C_{14} \cdot \psi_d^{(RU)}}{T_{14}}\right) \quad (6)$$

where $\delta_{DU} \in [0, 1]$ represents the fraction of the sub-component mapping shared by different O-RAN elements, and $\psi_d^{(RU)}$ is the number of RUs connected to the DU $d \in \{1, \dots, n_{DU}\}$.

For the CU elements, the power consumption of the CU $c \in \{1, \dots, n_{CU}\}$ must be computed in a different way. Considering that the CU processes data in terms of packets, we need to express the power consumption considering the incoming workload. In this way, we can define the power consumption of the CU $c \in \{1, \dots, n_{CU}\}$ as follows:

$$\bar{\Phi}_{CU}(c, \lambda) = \bar{\Phi}_{CU}^{(idle)}(c) + \lambda s_{act} [\bar{\Phi}_{CU}^{(act)}(c) - \bar{\Phi}_{CU}^{(idle)}(c)] \quad (7)$$

where λ is the normalized incoming workload ($\lambda \in [0, 1]$) and $s_{act} > 0$ is a scaling parameter. It is important to remember that it is possible to express the power consumption $\bar{\Phi}_{DU}(d)$ in the same way as done in (7) for CUs. Nevertheless, considering that the average incoming workload over a long-term time window to the DU remains constant (or, at most,

the variations are insignificant), the definition in (6) can be a reasonable approximation. Instead, $\tilde{\Phi}_{CU}^{(act)}(c)$ and $\tilde{\Phi}_{CU}^{(idle)}(c)$ represent the power consumption during the active and idle periods, respectively. During the active periods, the CU is processing packets, while during the idles ones, it is waiting for packets to process.

To compute these terms, it is needed to map the implementation of the logical CU element into the physical entities. To be more precise, we must compute the number of physical entities to satisfy the processing capacity that the logical elements must support. The CU $c \in \{1, \dots, n_{CU}\}$ must support the incoming workload from the $\psi_c^{(DU)}$ DU elements. Without loss of generality, we can express the number of chosen physical instances required for the CU $c \in \{1, \dots, n_{CU}\}$ by using the following equation:

$$\omega_{CU}(c) = \frac{\psi_c^{(DU)} \cdot (C_{14} + C_{15})}{\mu \cdot \Phi_{phy} \cdot T_{phy}} \quad (8)$$

where $\mu \in (0, 1]$, Φ_{phy} , and T_{phy} are the peak supported normalized processing capacity, max power consumption, and technology-dependent parameter of the chosen physical instance, respectively. The denominator in (8) is the max supported processing capacity of a single physical entity. As a typical example, the physical instance is a COTS Central Processing Unit (CPU) with the DU implemented as a datacenter of servers. In this example $\omega_{CU}(c)$ represents the number of CPUs to satisfy the workload to the CU $c \in \{1, \dots, n_{CU}\}$ incoming from the $\psi_c^{(DU)}$ DUs. They are expressed in (7) and (8). Finally,

$$\tilde{\Phi}_{CU}^{(act)}(c) = \omega_{CU}(c) \cdot \psi_c^{(DU)} \cdot \Phi_{phy} \quad (9)$$

$$\tilde{\Phi}_{CU}^{(idle)}(c) = \theta_{idle} \cdot \tilde{\Phi}_{CU}^{(act)}(c) \quad (10)$$

where $\theta_{idle} \in (0, 1)$ indicates the fraction of the active power consumption in the idle periods.

To capture the effect of a workload profile, we express $\bar{\Phi}_{CU}(c)$ as the expectation of $\tilde{\Phi}_{CU}(c, \lambda)$ with respect to the incoming workload λ :

$$\bar{\Phi}_{CU}(c) = \int_0^1 \tilde{\Phi}_{CU}(c, \lambda) \cdot \rho_\lambda(\lambda) d\lambda \quad (11)$$

where $\rho_\lambda(\lambda)$ is the probability density of the incoming workload from the connected DUs. Instead, to calculate the power consumption of the interfaces, we should consider the different ones used in the RAN and O-RAN architectures. In the RAN architecture, we have two interfaces: fronthaul between the RRH and the BBU and backhaul between the BBU and the core network; whereas, in the O-RAN architecture, we have an additional interface (midhaul) between the DU and the CU. In this way, we can express the power consumption of the various interfaces as Φ_{FH} , Φ_{MH} and Φ_{BH} for fronthaul, midhaul and backhaul, respectively.

The power consumption of common site infrastructure Φ_{CSI} consists of the power consumed for cooling, monitoring, and lights. It is not dependent on different processes in nodes of RAN and O-RAN architectures, or their sub-components, and a constant value can be considered for it. For the RAN

case, Φ_{CSI} is present in the BBU, while for the O-RAN case, it is present in DU and CU contributions.

Based on the power consumption of the three main components, we can calculate the power consumption of the different elements of the two architectures.

For the RAN case, the power consumption of the RRH and BBU elements can be expressed as:

$$\Phi_{RRH} = \tilde{\Phi}_{RRH} + \Phi_{FH} \quad (12)$$

$$\Phi_{BBU} = \tilde{\Phi}_{BBU} + \Phi_{BH} \quad (13)$$

Instead, for the O-RAN case, the power consumption is expressed as:

$$\Phi_{RU} = \tilde{\Phi}_{RU} + \Phi_{FH} \quad (14)$$

$$\Phi_{DU}(d) = \tilde{\Phi}_{DU}(d) + \Phi_{MH} + \Phi_{CSI} \quad \forall d \in \{1, \dots, n_{DU}\} \quad (15)$$

$$\Phi_{CU}(c, \lambda) = \tilde{\Phi}_{CU}(c, \lambda) + \Phi_{BH} + \Phi_{CSI} \quad \forall c \in \{1, \dots, n_{CU}\} \quad (16)$$

Hence, the power consumption of the two architectures is expressed as:

$$\Phi_{RAN} = n_{RRH} \cdot \Phi_{RRH} + n_{BBU} \cdot \Phi_{BBU} \quad (17)$$

$$\Phi_{O-RAN} = n_{RU} \cdot \Phi_{RU} + \sum_{d=1}^{n_{DU}} \Phi_{DU}(d) + \sum_{c=1}^{n_{CU}} \Phi_{CU}(c) \quad (18)$$

where n_{RRH} , n_{BBU} , n_{RU} , n_{DU} , and n_{CU} indicate the number of RRH, BBU, RU, DU, and CU elements of the RAN and O-RAN architectures considering a specific scenario.

Finally, the power percentage difference is defined with the following equation:

$$\Delta\Phi\% = \frac{\Phi_{RAN} - \Phi_{O-RAN}}{\Phi_{RAN}} \cdot 100 \quad (19)$$

B. Estimation of Edge Facility Placement

As we aim to create an architecture based on the pre-existing locations of RUs from earlier generations of RAN, the focus of this section is on DU/CU allocation using the 3-layer architecture (Fig. 1) for O-RAN. This monolithic architecture serves as a reference point for evaluating the power consumption aspects of O-RAN deployment, taking into account the potential challenges associated with a tightly integrated design. Examining the power consumption patterns in this worst-case scenario can provide valuable insights into the factors that may influence energy efficiency in more complex O-RAN configurations. Though the algorithm that we adopted DU/CU allocation is known in the literature as a one of the possible methods for network optimization, we note that the novelty of our approach lies in its application in conjunction with the metrics specified by our model. The latter is specific to the O-RAN context, and allows tuning various parameters and constraints pertaining to O-RAN networks.

The positioning of the various radio-network units is a difficult task. Network operators must regularly adjust their networks to accommodate expected aggregated-traffic changes, as well as the QoS demanded by user requests and the system itself. As a result, they must carefully plan the network to efficiently install and migrate (if necessary) virtual machines running DUs and CUs to different locations in the network, as well as construct light-paths in advance. As the RU

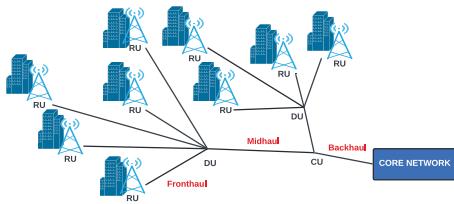


Fig. 1. DU/CU Placements in the 3-Layer O-RAN Architecture.

remains fixed at the cell site, mobile operators can place DUs and CUs on different metro aggregation nodes with the goal of minimising system power consumption. To achieve QoS and proper network operations, the allocation must ensure that both functional split and service latency limitations are respected for all network needs. Network operators need to consider the consumption of both the network and the processing elements and optimise the deployment to contain overall power consumption.

However, we note that the optimal placement of DUs and CUs may not always coincide with the existing fixed node locations, which may have constraints on hosting DUs in terms of security, energy, environmental, and other factors. Therefore, the use of existing fixed nodes may reduce the energy efficiency gains achieved by the proposed algorithm. Moreover, the construction of new nodes at the optimal locations may not be feasible for some networks due to cost and environmental impact. For example, in France and Ireland, the deployment of new nodes is restricted by the regulations on eCO₂ emissions.

Another challenge that may affect the optimal placement of DUs and CUs is the availability and cost of dark fiber to connect RUs and DUs. Dark fiber is not uniformly distributed across different network segments, and may require crossing multiple fixed networks (access networks, pre-aggregation networks, aggregation networks) with different conditions to deliver a dark fiber solution. Access network with Fiber to the Home (FTTH) is the most appropriate to propose a dark fiber solution, but an aggregation network is not so rich in terms of fiber (or in other words: not at the same price or availability or time-to-build). For instance, in Italy, Open Fiber is a company that aims to bring ultra-broadband to the whole country using FTTH technology, but it faces challenges in deploying dark fiber in some areas due to regulatory and technical barriers [35]. Thus, our algorithm should be seen as an optimistic upper bound on the energy efficiency that can be achieved by O-RAN deployment, and the actual performance may vary depending on the network conditions and constraints.

The proposed algorithm consists of two key steps. Firstly, it leverages the distribution pattern of RUs within a designated area to identify the minimum number of DUs required to ensure comprehensive coverage for all RUs. This step aims to optimize the DU placement while ensuring effective connectivity with the RUs. Secondly, based on the determined DUs from the previous step, the algorithm proceeds to select the minimum number of CUs necessary to cover all the identified

DUs. This step focuses on optimizing the CU placement to ensure efficient coordination and management of the DUs.

As previously mentioned, one of the primary challenges in transitioning from the previous two-layer architecture to the new three-layer architecture is the strategic placement of DUs and CUs while minimizing power consumption. If the RUs were uniformly distributed, this task would be relatively straightforward. However, the challenge lies in the significant variation of RU distribution across different regions. For instance, metropolitan areas have a considerably higher RU density compared to rural areas.

The objective at this stage is to group RUs together and assign a DU to each group. This task must adhere to certain constraints. Each DU has a maximum coverage range. This means that the farthest RU in a group should not exceed this coverage range from its associated DU. Moreover, each DU has a limited capacity to cover a certain number of RUs, which the placement algorithm must also take into consideration.

Different algorithms exist for the purpose of grouping objects using a similarity function. These algorithms, referred to as clustering algorithms, vary in their effectiveness depending on the specific problem. In our particular scenario, we have data that includes the locations of various RUs spread across a particular region. The objective is to group these RUs together based on the aforementioned constraints and find a point to locate the DU which has the minimum distance to all the RUs in that cluster. To accomplish this, we employ the k-means algorithm, originally introduced by Lloyd [36] and widely utilized in data mining, particularly in the domain of unsupervised learning.

k-means is a clustering algorithm with computational complexity of $\mathcal{O}(n^{d \cdot k + 1})$, where n is the number of objects, d is the dimension of properties, and k is the number of clusters. In fact, the computational complexity of the k-means algorithm positions it as a highly suitable and advantageous choice for solving optimisation problems. The efficiency of the k-means algorithm arises from its ability to swiftly converge towards optimal solutions, even when dealing with large datasets.

The clustering results in this method may be obtained by optimising an objective function. In our specific case, where the criterion revolves around the distance between objects, the objective function is structured to minimise this distance. The primary goal of the clustering process is to identify clusters where the overall distance between objects within each cluster is kept to a minimum.

Suppose to have n RUs $\{x_1, x_2, \dots, x_n\}$ with 3 dimensions which are their geographical coordinates and that should be divided into k clusters. We define these clusters by a set called $S = \{S_1, S_2, \dots, S_k\}$. The members of the clusters must be selected from RUs in a way that minimises the Within-Cluster Sum of Squares (WCSS). Therefore, the objective function is as follows:

$$\arg \min_s \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 \quad (20)$$

where μ_i is the mean of points in cluster S_i . Lloyd's algorithm is used as a standard for k-means clustering, especially in the field of computer science. We will first refer to the notations

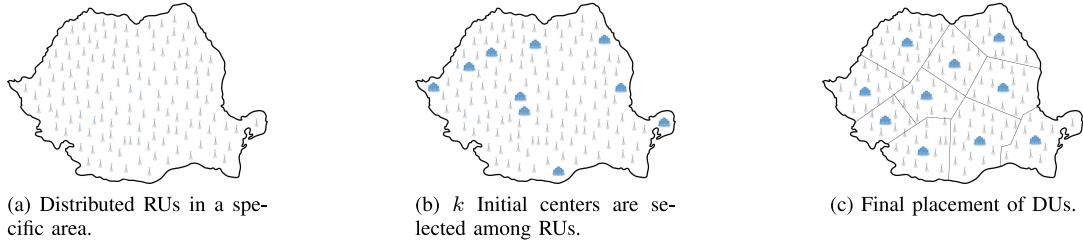


Fig. 2. Placement of DUs based on the proposed algorithm.

used in this regard. We define $m_i^{(t)}$ to represent the average of all data points that belong to the i^{th} cluster in the t^{th} iteration of the algorithm and $S_i^{(t)}$ to represent the set of members of the i^{th} cluster in the t^{th} iteration. We should also consider the first initial k centers that are selected among input data, called $m_1^{(1)}, m_2^{(1)}, \dots, m_k^{(1)}$.

Lloyd's algorithm can be divided into two steps according to the above symbols, which are:

- 1) Assignment step, and
- 2) Update step.

Assignment step: Assign each RU to the nearest cluster. This means that the Euclidean distance of each RU from the centers is measured, and that RU will be a member of the cluster that has the least Euclidean distance from the center of the cluster.

$$S_i^{(erVar)} = \left\{ x_p : \eta(x_p, m_i^{(erVar)})^2 \leq \eta(x_p, m_j^{(erVar)})^2 \quad \forall j \quad 1 \leq j \leq k \right\} \quad (21)$$

where $\eta(a, b)$ represents the Euclidean distance between the points a and b .

Update step: The means of new clusters are calculated.

$$m_i^{(erVar+1)} = \frac{1}{|S_i^{(erVar)}|} \cdot \sum_{x_j \in S_i^{(erVar)}} x_j \quad (22)$$

This algorithm iterates these two steps until the membership of any RU is unchanged. The final center points of each cluster are the proposed locations for DUs. A main deficiency of Lloyd's algorithm is that k objects are randomly selected as centroids in the initial step, which in later steps are changed to the average value of each cluster. k -means++ introduces a better method for initialisation of these k centroids which also increases the speed of the algorithm.

Selection of the initial number of clusters, k , in the k -means algorithm is crucial for achieving optimal placement results and algorithmic efficiency. In our proposed method, the DUs have a maximum coverage range, representing the farthest distance they can support to cover an RU. This coverage range implies that each DU can cover RUs within a circular space with a radius equal to its coverage range. To determine the initial value for k , we divide the total area where RUs are distributed by the surface area of the circular space defined by the DU's coverage range. This calculated value of k serves as the starting point for the k -means algorithm, where the initial DU locations are determined among the RUs. However, there are additional constraints that need to be considered beyond

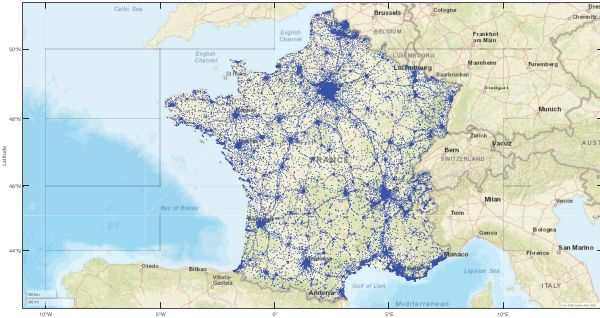
what the k -means algorithm accounts for. To ensure that the constraints are met, the distance between each RU and its associated DU is calculated, verifying that it does not exceed the maximum distance a DU can support. Additionally, the number of RUs connected to each DU should not surpass the maximum capacity of nodes that a DU can handle. If any of these constraints are violated, the affected cluster undergoes re-clustering using the k -means algorithm, by assigning an additional centroid to address the constraint violation. This process is applied to all nodes in the network to ensure that all the constraints are satisfied and the optimal placement is achieved.

For better understanding, suppose that we have a map of distributed RUs in a specific area, as depicted in Fig. 2a. Based on the proposed algorithm, k initial centers are selected among input data, as depicted in Fig. 2b. Suppose that in this example k is equal to 10. Then, RUs are assigned to their nearest centroids and iterations of the algorithm start, until no change happens in the membership of any RU. The final result is illustrated in Fig. 2c. Now all clusters will be evaluated for any aforementioned constraints and if necessary re-clustering may occur.

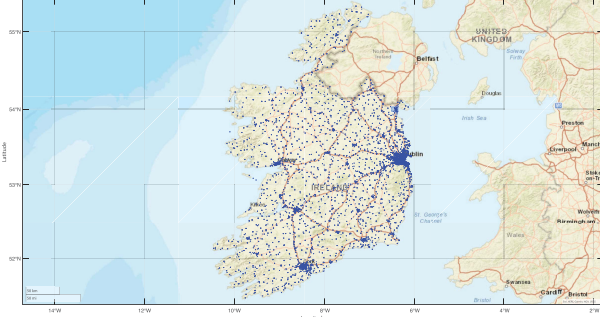
Following the identification of the DUs, the subsequent phase involves determining the optimal placement of CUs based on the already established DUs. This process is executed in a manner similar to the earlier step, which involved locating DUs over the RUs. Similar to DUs, each CU has specific considerations such as a maximum coverage range and a maximum number of connected DUs. These factors need to be taken into account during the placement process. The objective is to ensure that CUs are strategically positioned to cover the desired geographical area where the RUs are distributed. By applying similar methodologies as before, the algorithm determines the optimal locations for both DUs and CUs within the given geographical area. The final results of this phase comprise the identified placements for DUs and CUs, offering a comprehensive solution for the optimal positioning of units in the network.

IV. NUMERICAL RESULTS

This section focuses on analyzing the power consumption of a base O-RAN architecture that was previously generated using the deployed RAN's architecture. To achieve this, the proposed power consumption model is applied to a real-world implementation of O-RAN, specifically examining the datasets of RUs deployed in France and Ireland (see Fig. 3). We



(a) France.



(b) Ireland.

Fig. 3. Distribution of RUs in France (Fig. 3a) and Ireland (Fig. 3b).

chose these two countries as the locations for our experiment because they are among the few countries that have data on O-RAN networks and power consumption provided by the European Commission Joint Research Centre (EC JRC) [37]. We acknowledge the valuable contribution of the EC JRC and their partners in collecting and sharing these datasets. We recognize that these locations may not represent the diversity and complexity of real-world scenarios in other regions. However, our main goal in this paper is to propose a power consumption model for O-RAN networks, and we just used these datasets to validate our model and show its applicability for future research. By utilizing the model, we aim to gain insights into the energy requirements and consumption patterns within this particular O-RAN implementation. The analysis of power usage in this base architecture adds to our understanding of energy efficiency in O-RAN networks and provides useful information for reducing power usage and improving the architecture's sustainability.

In the first subsection, the proposed placement algorithm is applied to derive an optimal or near-optimal solution for the placement of DUs and CUs based on the provided dataset of RUs in France and Ireland. This algorithm takes into consideration various factors such as coverage requirements, connectivity constraints, and maximum capacity limitations of the units.

Moving on to the next subsection, the power consumption of the O-RAN implementation is calculated based on the previously determined locations of the DUs and CUs in France and Ireland. The proposed power consumption model is employed to estimate the power requirements associated with the deployed units.

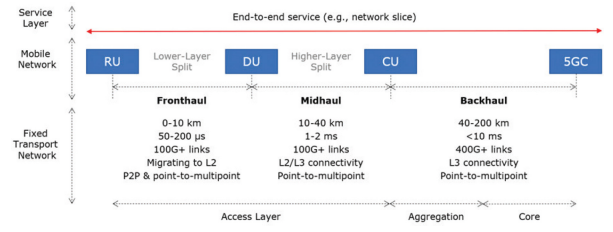
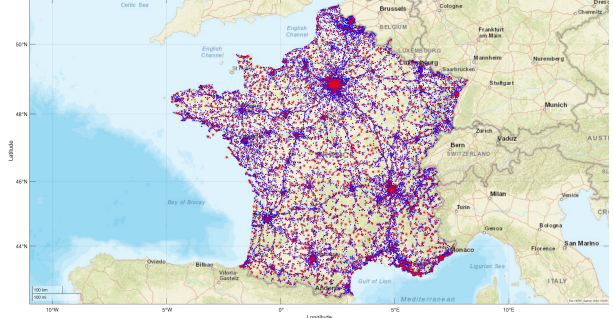


Fig. 4. Reference values for 5G O-RAN Splitting Elements.



(a) DUs over RUs



(b) CUs over DUs

Fig. 5. Distribution of CUs and DUs in France.

The reference values of the 5G O-RAN splitting elements are illustrated in Fig. 4. These values are used in the placement algorithm and the power consumption analysis described in the next subsections.

A. 5G O-RAN Elements Placement

The data of France consists of 42,797 RU nodes operated by different vendors across the country. In the first step, we consider that DU and CU nodes have the capability of collecting data of all operators together. The first task is to find the best placement for DUs in a way that supports all distributed RUs. For this job based on the proposed method, k -means will find k initial centroids and the initial value for k is calculated based on the proposed method. For France, the initial k is calculated at 1733. The maximum distance each DU supports is considered equal to 10 km. Fig. 5a illustrates the placement of DUs in France. Red stars demonstrate the location of DUs.

Using Fig. 4 as a reference, the distance between a DU and a RU can vary from 0 to 10 kilometers. To illustrate the impact of utilizing DUs with shorter maximum distances,

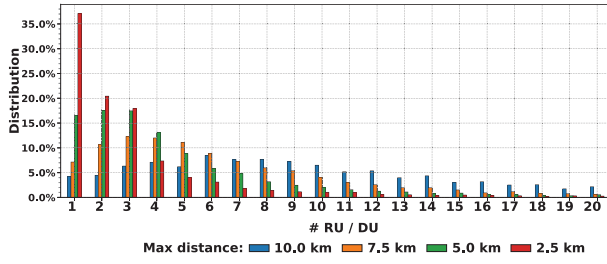


Fig. 6. Distribution of DUs over all RUs in France with different maximum covering distances for DUs.

we also evaluate other values within this range, specifically 7.5 km, 5 km, and 2.5 km. These distances may not reflect the real-world constraints and technological impairments that affect the communication latency between the RUs and the DUs. However, our main goal here is to propose a power consumption model for O-RAN networks, and we just used a simple placement algorithm with some feasible distances to show how this power model can be used for future research. Future work can focus on more realistic and complex scenarios based on our proposed power consumption model. The outcomes of these evaluations are visually represented in Fig. 6.

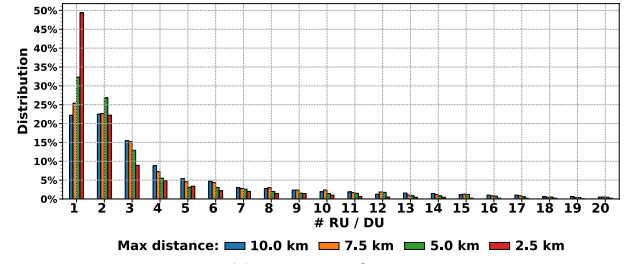
Based on the chart, as the maximum coverage distance of DUs decreases, the number of DUs that only cover one RU increases significantly and reaches over 35% for the case when the maximum distance is 2.5 km. Therefore, we can conclude that greater coverage distance for DUs leads to better clustering of RUs.

France has four major mobile operators based on the dataset; namely, *Bouyges Telecom* with 9,788 RUs, *Free Mobile* with 18,883 RUs, *Orange* with 6,026 RUs and *SFR* with 8,100 RUs. For the cases where operators cannot use a common DU infrastructure, tests are also run on the RUs of these operators separately. The results of tests on each operator's RUs with the different maximum supporting ranges for DUs are shown in Fig. 7.

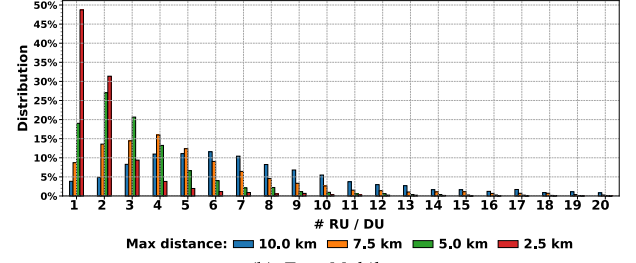
Based on the results, we can see that the percentage of DUs that support fewer RUs increases as the total number of RUs of different operators decreases. For instance, less than 5% of DUs support only one RU for the *Free Mobile* operator with 18883 RUs; however, this value increases to about 27% for the *Orange* operator with the lowest number of RUs, which is 6026. Additionally, the impact of decreasing the maximum coverage distance of DUs is also depicted in the charts, as we mentioned before.

After the placement of DUs, in the second phase, we try to find the best places for CUs over them. Fig. 5b depicts the distribution of CUs over DUs in France for the case when data of all operators can be collected together. Blue dots demonstrate the location of DUs from the first phase and red stars depict the places of CUs located over them.

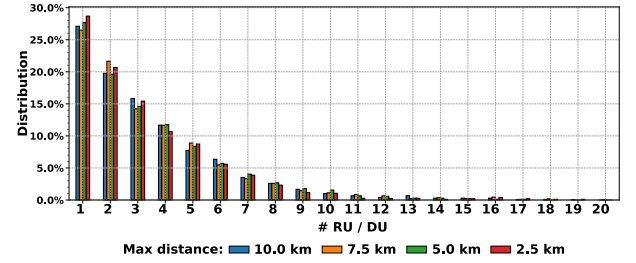
By referring to Fig. 4, the evaluation includes testing four different maximum supporting distances for CUs in the range of 10 to 40 kilometers. The specific distances examined are 40 km, 30 km, 20 km, and 10 km. The outcomes of these tests are visually presented in Fig. 8, showing the results of the



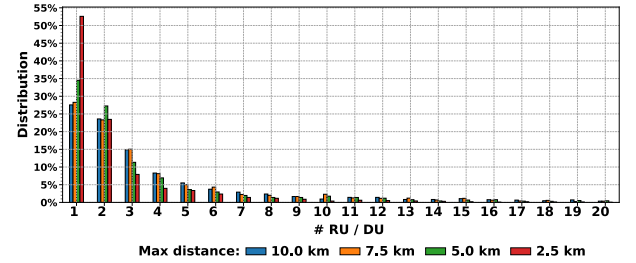
(a) *Bouyges Telecom*



(b) *Free Mobile*



(c) *Orange*



(d) *SFR*

Fig. 7. Distribution of DUs over RUs of *Bouyges Telecom* (7a), *Free Mobile* (7b), *Orange* (7c), and *SFR* (7d) with different maximum covering distances for DUs.

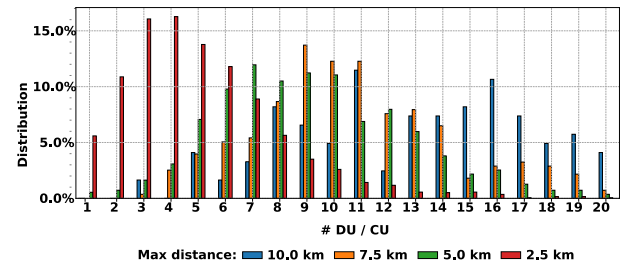


Fig. 8. Distribution of CUs over all DUs in France with different maximum covering distances for CUs.

power consumption analysis for each respective CU distance setting.

As mentioned before, we can see that the best results happen on the greatest distances each CU supports, whereas the worst

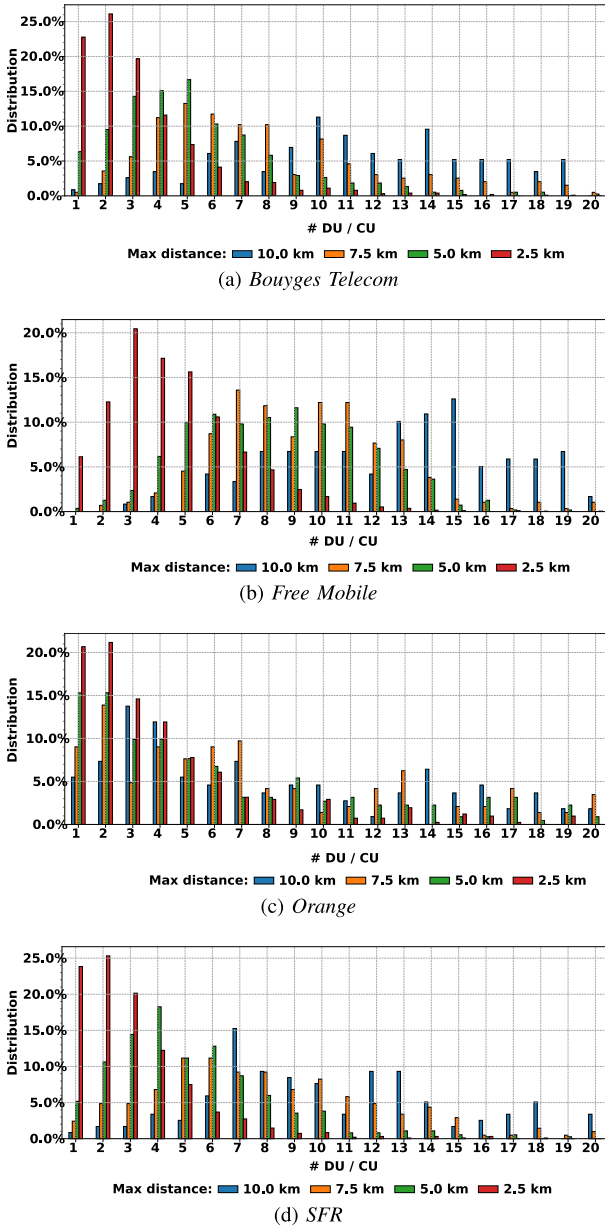
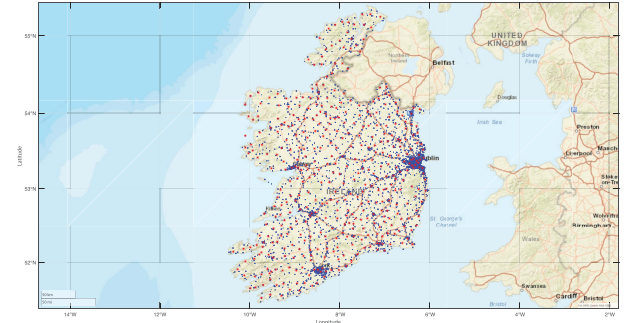


Fig. 9. Distribution of CUs over DUs of *Bouyges Telecom* (9a), *Free Mobile* (9b), *Orange* (9c), and *SFR* (9d) with different maximum covering distances for CUs.

case is for the lowest maximum distance, which is 10 km. We also ran the tests on the operators separately to compare the results for different operators. The results of these tests are shown in Fig. 9.

Once again, the analysis reveals that the best results are obtained when an operator deploys a larger number of RUs and increases the maximum supporting distance for the CUs. These circumstances consistently produce the best outcomes across all test cases.

The dataset of Ireland consists of 10,198 points with three major operators; namely: *Vodafone*, *Three*, and *Meteor* with 2,476, 5,137, and 2,585 RUs, respectively. Fig. 10 illustrates the graphical mapping of DUs over RUs with a maximum coverage distance of 10 km for each DU. For Ireland, the



(a) DUs over RUs



(b) CUs over DUs

Fig. 10. Distribution of CUs and DUs in Ireland.

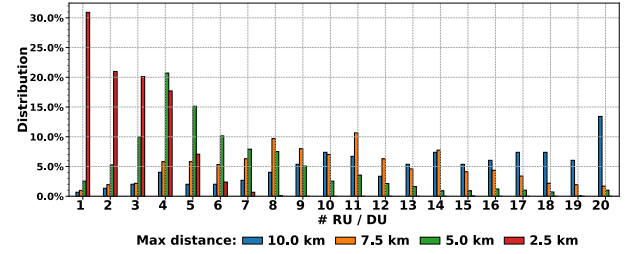


Fig. 11. Distribution of DUs over all RUs in Ireland with different maximum covering distances for DUs.

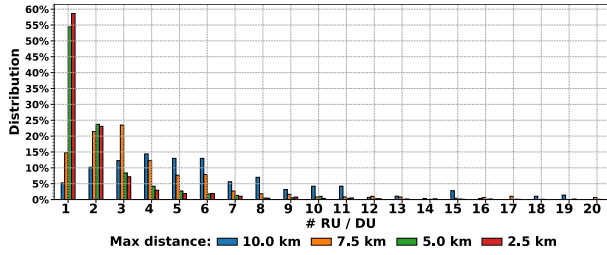
initially calculated k for running the algorithm based on the proposed method is 269.

In the first phase, the placement is done to find the best locations for DUs over RUs. Similar to previous tests, we consider four maximum coverage distances for DUs (10km, 7.5km, 5km and 2.5km). Fig. 11 illustrates the result of the test on RUs of all operators together.

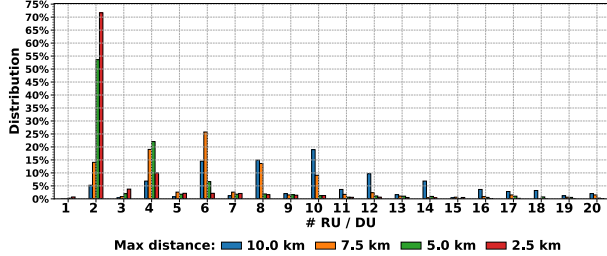
The impact of maximum distances for DUs is again visible in this case. When this distance is 10 km, we can see that most of the DUs cover more than 10 and up to 20 RUs, and only a small percentage of them cover areas with less than 9 RUs. On the contrary, if DUs cover only 2.5 km, more than 30% of RUs will be allocated to only one DU.

In the next step, the placement has been done on RUs of *Vodafone* and *Three* operators separately with different coverage distances for DUs. Results are illustrated in Fig. 12.

In the second phase, based on the placement of DUs in the previous step, the best placement is sought for CUs over them. If we consider all RUs together, the graphical result and the distribution of CUs over DUs are illustrated in Fig. 10b.



(a) Vodafone



(b) Three

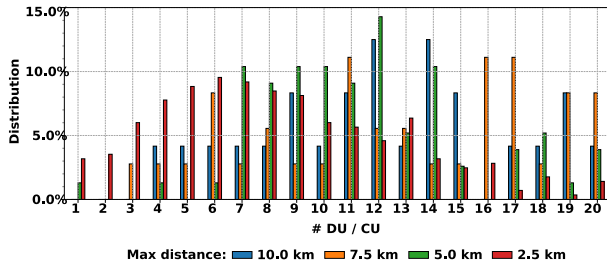
Fig. 12. Distribution of DUs over RUs of *Vodafone* and *Three* in Ireland with different maximum covering distances for DUs.

Fig. 13. Distribution of CUs over all DUs in Ireland with different maximum covering distances for CUs.

The great impact of higher maximum coverage distance for CUs can be seen in Fig. 13. Setting the distance to 40 km, there is no CU covering less than 4 DUs and most of them support between 9 to 15 DUs, whereas if the maximum coverage distance decreases to 10 km the percentage of CUs supporting only few DUs increases significantly.

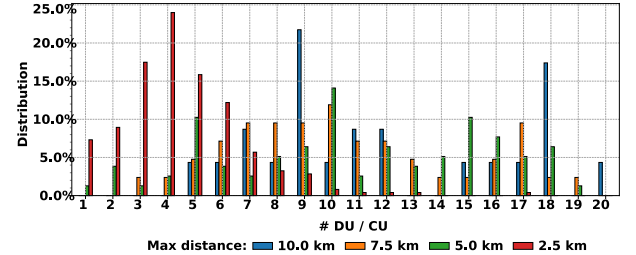
Finally, Fig. 14 illustrates the results of placement of CUs over DUs of *Vodafone* and *Three* operators separately.

B. Power Consumption

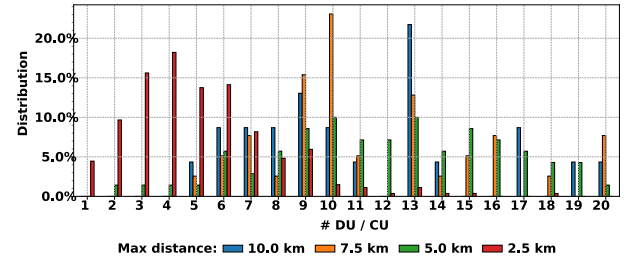
Considering the architectures of RAN and O-RAN specifications, the possible values of the reference complexity C_i are listed in Table III, based on the different processing operations (downlink or uplink) [34].

The possible values of the technology-dependent parameter T_i depend on the specific hardware (or software) implementation of the sub-component $i \in I$. Some examples can be found in [21], [22].

Regarding the technologies used for each sub-components of the RAN and O-RAN elements, we consider the case where the CU is realised with COTS hardware, while the other elements are developed using dedicated hardware (Application Specific Integrated Circuit (ASIC) in this case).



(a) Vodafone



(b) Three

Fig. 14. Distribution of CUs over DUs of *Vodafone* and *Three* in Ireland with different maximum covering distances for CUs.

TABLE III
REFERENCE COMPLEXITY OF DIGITAL COMPONENTS

Sub-component	Downlink [GOPS]	Uplink [GOPS]
Predistortion	10.7	0
Filtering	6.7	6.7
Up/Down-sampling	2	2
TD estimation/compensation of non-idealities	1.3	6.7
FFT/IFFT, FD non ideal	4	4
MIMO precoding	1.3	0
Synchronization	0	2
Channel estimation & interpolation	0	3.3
Equalizer computation	0	3.3
Equalization	0	2
OFDM Modulation/Demodulation	1.3	2.7
Mapping/Demapping	1.3	2.7
Channel Coding	1.3	8
Platform Control	2.7	1
Network Processing	8	5.3

Based on the placement of DUs and CUs in France and Ireland, we can now calculate the total power consumption of these networks. To evaluate the results, we consider the reference scenario where the RU (and the RRH for the corresponding RAN case) operates at 20 MHz bandwidth, fully loaded system, Single-Input and Single-Output (SISO) antenna configuration, spectral efficiency 6 bit/s/Hz, and 64 Quadrature Amplitude Modulation (QAM) modulation scheme with coding rate equal to 1.

In accordance with [38], for the interfaces between the elements, we consider the power consumption values of 18.2 W for the fronthaul (Φ_{FH}), 10 W for the midhaul (Φ_{MH}), and 1 W for the backhaul (Φ_{BH}). Additionally, based on the insights provided in the same reference, the power consumption for the Channel State Information CSI (Φ_{CSI}) is set at 2.1 kW.

For what concerns the parameters of the proposed power model we defined different ESs as shown in Table IV.

TABLE IV
EVALUATION SCENARIOS AND PARAMETER VALUES OF
THE PROPOSED POWER MODEL

Parameter	α_i	β_i	γ_i	δ_{DU}	s_{act}	θ_{idle}	μ
ES(A)	1.5	0.0	1.5	0.4	1.5	0.3	0.7
ES(B)	2.0	0.0	1.5	0.4	1.5	0.3	0.7
ES(C)	2.0	0.0	0.5	0.4	1.5	0.3	0.7

For the probability density of the incoming workload $P_\lambda(\lambda)$, we use the results for the normalised workload during working days obtained in [39]. Finally, for the Φ_{phy} and T_{phy} parameters, we consider COTS hardware. Hence, $T_{phy} = T_{CPU} = 2$ and $\Phi_{phy} = 85 \text{ W}$.²

Using the proposed model, we compute the power consumption and the percentage difference between O-RAN and RAN in the different ESs. For each evaluation, we consider different values of the maximum distance supported between RUs and DUs (10km, 7.5km, 5km and 2.5km), and between DUs and CUs (40km, 30km, 20km and 10km), that we define as sets MD(1)-MD(4).

The results for ES(A) are shown from Figure 15a to Figure 15d. The first two figures correspond to the France scenario, while the last two relate to the Ireland scenario. With France operator data and MD(1), MD(2), and MD(3) sets, the power consumption between the RAN and O-RAN cases is almost similar. Instead, with MD(4), the O-RAN architecture consumes 20% more than the RAN one.

This behaviour is even more pronounced with Ireland operator data. As shown in Figures 15c and 15d, with MD(4) set, the O-RAN architecture is almost the triple of the RAN case.

In ES(B), we increase the parameter α_i from 1.5 to 2. The results are shown Figure 16a to Figure 16d. With this configuration of the parameters, the power consumption difference is lower than in ES(A). In this case, the O-RAN architecture has a power consumption 5% higher than the RAN case. Even for less stringent values of maximum distance supported (MD(1), MD(2), and MD(3)) the O-RAN architecture has a lower power consumption.

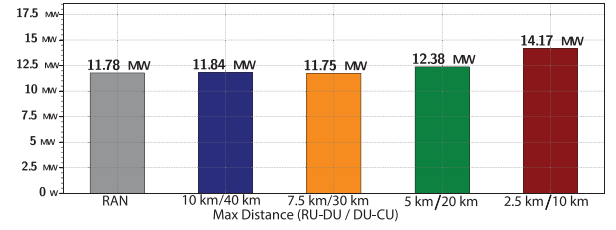
As shown in Figures 16c and 16d, also with the Ireland operator data there is an improvement with a lower power difference compared to the ES(A).

Finally, in ES(C), we decrease the parameter γ_i from 1.5 to 0.5. The results are shown Figure 17a to Figure 17d. This parameter configuration leads to an increase in power consumption compared to ES(A) and ES(B).

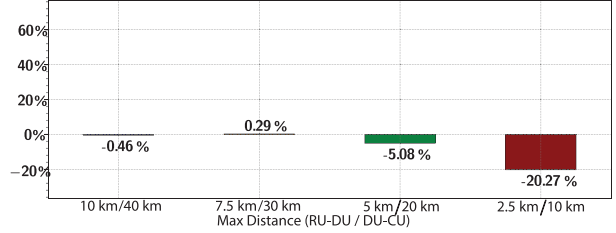
The case with Ireland operator data is the most significant one. In the case of the most stringent values of maximum supported distance, the O-RAN architecture has almost 8 times greater consumption than the RAN one (see Figure 17d).

C. Discussion

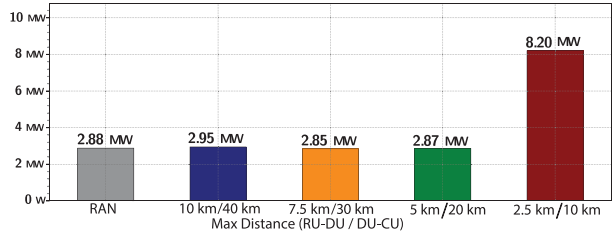
The findings from our evaluation of power consumption in the base O-RAN implementation have revealed that various parameters significantly impact the overall energy usage. The



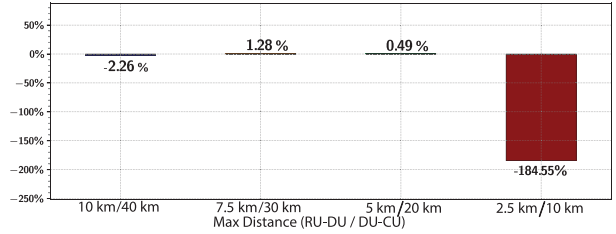
(a) Power consumption for France operator data.



(b) Power percentage difference for France operator data.



(c) Power consumption for Ireland operator data.

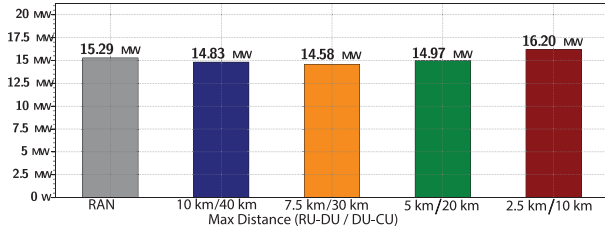


(d) Power percentage difference for Ireland operator data.

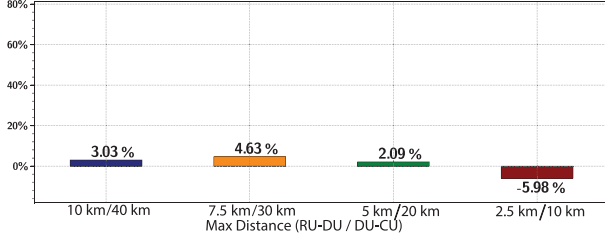
Fig. 15. Power consumption and percentage difference of RAN and O-RAN according to (19) for France and Ireland in ES(A).

role of softwarization is one important factor that significantly affects power consumption in O-RAN architectures. The development methodology for software elements becomes essential in determining the network's overall energy efficiency, because O-RAN heavily relies on software-based functionalities [40]. Energy consumption can be significantly reduced by implementing effective software strategies and enhancing algorithms. Adopting intelligent and adaptive software algorithms, for instance, can optimize resource allocation, improving load balancing and reducing irrational network energy consumption. The principles of Software Defined Networking (SDN), which enable dynamic and fine-grained control over routing and forwarding decisions, are also essential for managing the network's infrastructure. Because of their increased ability to adapt network operations to changing demand, operators can better utilize available resources and reduce energy consumption. Furthermore, the use of AI and machine learning algorithms in network management can have

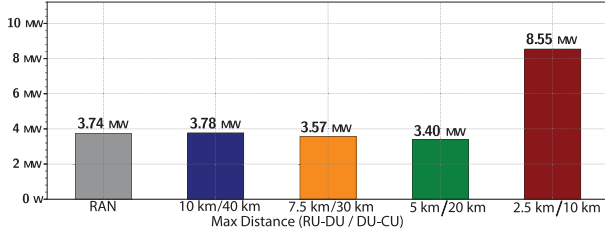
²<https://www.buildcomputers.net/power-consumption-of-pc-components.html>



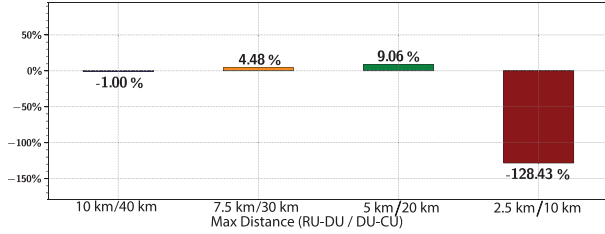
(a) Power consumption for France operator data.



(b) Power percentage difference for France operator data.



(c) Power consumption for Ireland operator data.



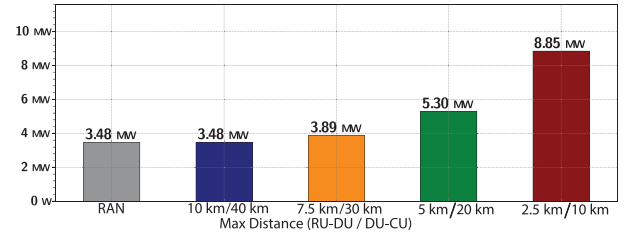
(d) Power percentage difference for Ireland operator data.

Fig. 16. Power consumption and percentage difference of RAN and O-RAN according to (19) for France and Ireland in ES(B).

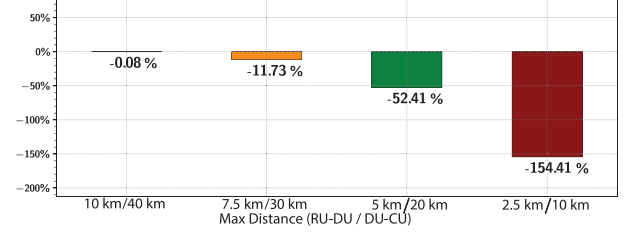
a significant positive impact on energy efficiency. AI-driven solutions can forecast demand fluctuations and take proactive actions to adapt network resources accordingly by continuously learning from network patterns and user behavior. This leads to energy savings.

Additionally, two other significant factors affecting power consumption are the density of RUs and the longest distance each DU and CU can support. Longer distances and higher densities are typically associated with higher energy needs. As a result, carefully optimizing the positioning and arrangement of DUs and CUs can significantly reduce power consumption.

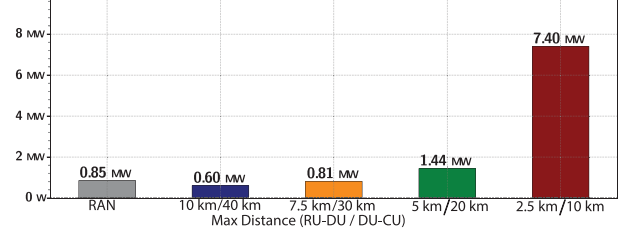
In the quest for energy-efficient O-RAN networks, accelerators have emerged as a key component in improving overall system performance while reducing power consumption in some situations. Specific computational tasks can be offloaded from general-purpose processors by means of specialized hardware accelerators, allowing for quicker and more energy-efficient completion of these tasks.



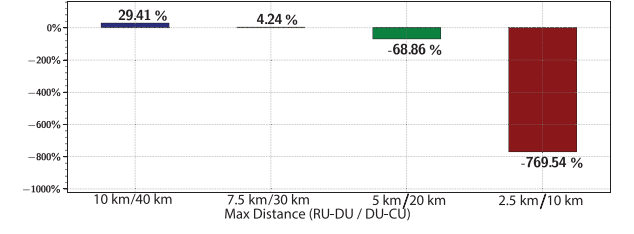
(a) Power consumption for France operator data.



(b) Power percentage difference for France operator data.



(c) Power consumption for Ireland operator data.



(d) Power percentage difference for Ireland operator data.

Fig. 17. Power consumption and percentage difference of RAN and O-RAN according to (19) for France and Ireland in ES(C).

V. CONCLUSION

This paper focuses on understanding energy consumption patterns in the 3-Layer O-RAN architecture. The primary goal is achieved by introducing a comprehensive power consumption model, implemented for O-RAN configurations in France and Ireland. The proposed model is used to calculate power consumption, allowing an exploration of parameter effects across different ESs. The research provides insights into energy consumption dynamics and optimization techniques within the O-RAN framework. Results show comparable power consumption to RAN in ES(A), with slight differences. However, in ES(B) and ES(C), power consumption increases, particularly in ES(C). The study offers valuable tools for researchers to explore optimization strategies for energy efficiency in O-RAN networks using the proposed model.

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